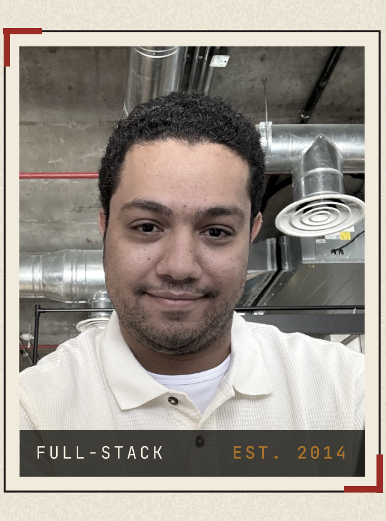
NO. 19 · VOL. FULL STACK   CURRICULUM VITÆ — THE FULL-STACK RECORD   EST. MMXIV

Ahmad Abdul-Jawad *Mahmoud*



FULL-STACK ENGINEER / BACKEND · WEB · MOBILE

CAIRO → RIYADH · DATABASE TO APP STORE, END TO END

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● SEE THE LIVE INTERACTIVE EDITION ↗

Full-stack engineer who owns a feature from the database to the app on the phone. Right now I am building a production ride-hailing platform **solo, across every layer**: a NestJS / TypeScript backend on PostgreSQL and Redis — **23 modules, ~320 endpoints, 2,900+ tests** — a Next.js operations console already in production, realtime Socket.IO feeds, payments, and native iOS & Android apps on one API contract. Behind it are **12+ years** of production engineering, seven of them owning the architecture of a POS used by 20k+ restaurants — so the APIs I design are the ones I would want to consume.

4	320+	2,900+	12+
PLATFORM LAYERS BUILT SOLO	API ENDPOINTS DESIGNED & SHIPPED	BACKEND TESTS I WROTE	YEARS IN PRODUCTION ENGINEERING

§ 01 Across the Stack

01 — BACKEND & DATA

Correct before clever

A NestJS modular monolith (**23 modules**) on PostgreSQL with versioned migrations; Redis for dispatch state, the driver geo index, rate limits and cron leases. A **double-entry ledger**, a transactional outbox and idempotency keys, so money and side effects survive retries.

02 — REALTIME & INTEGRATIONS

The parts that talk to the world

Socket.IO gateways for riders, drivers and admins; Google Maps routing with traffic-aware fare quotes and caching; Paymob cards, Apple Pay and Google Pay; WhatsApp-first OTP with SMS fallback; APNs and FCM push.

03 — WEB

An ops console people can run a business on

A **30-screen** Next.js 14 console: live MapLibre trips map, dispatch and SOS boards, KYC review, finance reconciliation, refunds and payouts, a map zone editor, feature flags, **RBAC** and an audit log behind Google SSO.

04 — MOBILE CLIENTS

12 years of shipping apps

Native SwiftUI (Swift 6) and Kotlin / Compose apps on shared, versioned core libraries, with **800+ iOS tests**. Before this: apps used by **35M+ people** — HungerStation, Ego ride-hailing and the Marn POS.

§ 02 Professional Experience

Mar 2026 — Present
CURRENT
INDEPENDENT

Full-Stack Engineer — Backend · Web · iOS · Android

Ride-Hailing Platform · Independent Project

Took the **full technical scope of a founding CTO** and carried it alone: chose the stack, set the standards, and have been building a production ride-hailing platform — a hardened backend and a real-time ops console **in production**, with native **iOS/SwiftUI** and **Android/Compose** apps (Rider & Driver on each) in active development. Every layer from data model to release, governed by explicit rules: idempotent mutations, double-entry accounting, acyclic module boundaries, and a CI test suite that gates the merge.

Backend APIs
NestJS · TypeScript · Mar '26 → now

Modular monolith, **23 modules** — **trip state machine**, **scored dispatch** over a Redis geo index, **double-entry ledger**, realtime rider/driver/admin **sockets**, JWT + phone-OTP, idempotency & audit log · ~320 endpoints over 47 controllers, traffic-aware fare quotes, **2,900+ tests**, Swagger docs.

Admin Console
Next.js 14 · Jun '26 → now

Bespoke **30-screen** ops surface — live **MapLibre** trips map, dispatch/SOS live boards, KYC onboarding review, finance reconciliation, refunds/payouts, cash control, map **zone editor**, feature flags, audit log, **Google SSO**.

iOS Apps · Rider & Driver
SwiftUI · Jul '26 → now

Two native **SwiftUI** apps on a shared **BuraqCore** Swift package (domain · networking · design system, pinned remotely) — live GPS **trip lifecycle**, Socket.IO feeds with fresh-token reconnect, in-app chat, payments, Live Activities, driver earnings & turn-by-turn navigation · Swift 6 strict concurrency, **800+ tests**.

Android Apps · Rider & Driver
Kotlin · Jetpack Compose · Sep '26 → now

Native **Kotlin / Compose** ports of both apps against **the same API contract**, on a shared **buraq-core-android** library published to GitHub Packages — driver KYC onboarding, offers and trips; rider booking, live tracking, FCM push and Google Pay · **290+ test suites**.

One backend · one contract · four native clients in flight — plus growth loops (promotions, referrals, loyalty tiers, wallet), driver safety/SOS, and scheduled reconciliation jobs. The exercise a technical leader needs periodically: own every trade-off yourself, then know exactly what you are asking of a team.

Aug 2019 — Aug 2026
RIYADH, KSA

iOS Team Lead – 7 years leading the iOS org

Marn — Jahez Group ↗

- **Led a 5+ engineer iOS team** for seven years — onboarding, mentoring, career growth, and the review culture behind an enterprise POS used by 20k+ restaurants and cafés.
- **Owned the technical roadmap and architecture:** an offline-first CoreData core with cloud sync, built so a venue keeps trading when the network does not.
- **Set the engineering standards** — module boundaries, code review, automated testing — that held the app at a 99.8–99.9% crash-free rate in production.
- **Drove the multi-year migration** to SwiftUI, Combine and async/await without pausing feature delivery, and made the case for it to product leadership.
- **Built CI/CD and the release process from scratch**, cutting deployment time 60% and shortening every engineer's feedback loop.
- Ran the performance programme (40% faster load times) and designed the table-management, order-processing and inventory systems at the heart of the product.
- Partnered with product, design and backend leadership on scope, estimates and delivery commitments the team could keep.

Dec 2025 — Jun 2026
VOLUNTEER

Senior iOS Engineer

HaramBlur ↗ · Voluntary Charity Project

- Designed an architecture integrating on-device machine learning for real-time content analysis.
- Built a seamless communication layer between native iOS and web-based content.
- Implemented efficient image processing with region-based and full-image blur effects.
- Created a real-time video monitoring system for continuous content filtering.
- Developed smart banner blocking and optimised memory management throughout.

Mar 2018 — May 2019
RIYADH, KSA

iOS Team Lead

Ego — The Ride Hailing App ↗

- Led iOS delivery for a complete ride-hailing ecosystem (rider, driver, watchOS) serving 15M+ users in Saudi Arabia.
- Implemented real-time GPS tracking and navigation with MapKit and optimised route algorithms.
- Integrated Apple Pay, HyperPay, and push-notification infrastructure.
- Created seamless iPhone ⟷ Apple Watch handoff for continuous ride tracking.
- Built the driver rating system and real-time fare calculation engine.

Sep 2016 — Mar 2018
KSA

Senior iOS Engineer

HungerStation — DeliveryHero Group ↗

- Contributed to a major food-delivery platform serving 20M+ users.
- Optimised app performance for high-volume transaction processing.
- Implemented real-time order tracking and delivery optimisation.
- Collaborated with backend teams on API design and integration.

May 2016 — Aug 2016

iOS Developer

ALNapedh ↗

- Developed food-ordering apps with Firebase integration and real-time features.
- Implemented Spreedly payments and live-chat functionality.
- Built responsive UI with Auto Layout across multiple device sizes.

Jul 2015 — Sep 2015

iOS & Apple Watch Developer

Nile Design Co. ↗

- Early adopter of Apple Watch development (watchOS 1–2).
- Built a foundation in iOS and watchOS app development.
- Adopted SwiftUI from WWDC 2019 onward.

May 2014 — May 2015

iOS Developer

UK Pro. Solutions Ltd. ↗

- Foundation iOS experience in Objective-C and early Swift.
- Built social networks and e-commerce platforms.
- Learned core iOS principles and App Store deployment.

§ 03 Selected Projects

01 — FULL-STACK

Ride-Hailing Platform (built from scratch · since Mar 2026)

SwiftUI · Kotlin · Jetpack Compose · NestJS · TypeScript · PostgreSQL · Redis · Socket.IO · Next.js

End-to-end ride-hailing platform architected solo: a hardened NestJS backend (trip state machine, scored dispatch, double-entry ledger, realtime sockets, 2,900+ tests) and a real-time Next.js operations console **in production**, with **four native clients** (Rider & Driver on iOS/SwiftUI and on Android/Kotlin·Compose) in active development.

◆ Four Platforms · Solo Ownership · Production Security

02 — ENTERPRISE

Marn — iPad POS

SwiftUI · CoreData · CloudKit · Real-time Sync

Enterprise point-of-sale serving 20k+ restaurants and cafés across Saudi Arabia. Offline-first architecture, complex table management, real-time order processing — **the product my iOS team owned for seven years**.

★ 4.8 / 5 — 2,437 ratings

03 — MOBILITY

Ego — Ride Hailing

MapKit · Core Location · Apple Pay · Push · watchOS

Complete ride-hailing ecosystem: rider app, driver app, and Apple Watch companion. Real-time GPS, Apple Pay, and cross-device experiences for 15M+ users.

★ 4.7 / 5 — 211.8k+ ratings

04 — DELIVERY

HungerStation

UIKit · Firebase · Real-time APIs · Perf Optimization

Major food-delivery platform serving 20M+ users across Saudi Arabia and Bahrain. Focused on performance and real-time order tracking.

★ 4.5 / 5 — 1.9M+ ratings

05 — AI / PRIVACY

HaramBlur — Privacy Browser

SwiftUI · CoreML · WKWebView · CoreImage · Vision

Privacy-focused iOS browser with real-time, AI-powered content detection and filtering. On-device ML, smart blur effects, and live video monitoring.

◆ AI-Powered Privacy Browser

06 — COMMUNITY

Daleelo

MapKit · Core Location · Social SDKs · Community

Muslim community app for users across the US — prayer times, mosque finder, and community features with enhanced location and social integration.

★ 4.5 / 5 — 1,287 ratings

§ 04 Stack & Craft

WEB & INFRASTRUCTURE FULL-STACK

- Next.js 14
- React
- TypeScript
- MapLibre
- Socket.IO
- Docker
- GitHub Actions CI
- Prometheus Metrics
- Jest
- Google Maps Platform
- Paymob
- Cloudflare R2

IOS — 2025 CORE

- Swift 6.0
- SwiftUI
- UIKit
- Combine
- async/await
- SwiftData
- iOS 18+

ANDROID — 2025 NATIVE

- Kotlin
- Jetpack Compose
- Coroutines
- Flow
- MVVM

ARCHITECTURE PATTERNS

- MVVM
- VIPER
- Clean Architecture
- Coordinator
- TDD

APPLE ECOSYSTEM PLATFORMS

- watchOS
- tvOS
- macOS
- WidgetKit
- Live Activities
- App Intents

MODERN FEATURES INTELLIGENCE

- Core ML
- Vision
- ARKit
- Dynamic Island
- App Tracking

BACKEND & APIS CORE

- NestJS
- TypeScript
- PostgreSQL
- TypeORM
- Redis
- JWT/OTP Auth
- REST
- GraphQL
- Firebase
- CloudKit
- WebSockets
- Swagger/OpenAPI

TOOLING & AI WORKFLOW

- Xcode 15+
- Fastlane
- TestFlight
- Instruments
- Git
- Copilot
- Cursor

§ 05 Portfolio Index

FOOD & GAMING

HungerStation — delivery
Punters — racing tips

TRANSPORTATION

Ego — Rider
Ego — Driver
Ego — Watch app

ENTERPRISE

Marn POS
Medical Guide

SOCIAL & COMMUNITY

Daleelo
NU-inc
VeeU

WATCHOS

Ego Watch
TaQs Weather
TweetMe

E-COMMERCE

iBss
Exclusive Luxe
Scratch

RELIGIOUS & EDU

Afasy
ArabNileGroup

UTILITY & TOOLS

Unlocking Smart
Insta
iContacts
🚀 *First iOS app*

AI & ML

HaramBlur — CoreML
RecognizeMe
🎓 *Grad project — OpenCV*

WEB

FaceDemo
Web Pages

§ 06 Record

EDUCATION

B.Sc. Information Systems

Computer Sciences College
Asyut University, Egypt · 2009–2013
iOS Semester — Excellent (91.5%)

CERTIFICATION

ISTQB Foundation Level

Egyptian Software Testing Qualification Board

LANGUAGES

Arabic	Native
English	Professional

§ 07 Honours

01 Building a **production ride-hailing platform solo across every layer** — backend, ops console, and iOS and Android apps on one API contract.

02 Held a **99.8–99.9% crash-free rate** and cut deployment time **60%** by building the standards, tests and CI/CD the team ran on.

03 Architecting a **full-stack ride-hailing platform solo** — NestJS/PostgreSQL backend and Next.js ops console in production, four native clients (rider and driver, iOS and Android) in development, with idempotent APIs and double-entry accounting.

04 Built a food-delivery platform serving **20M+ users** across Saudi Arabia and Bahrain, and a ride-hailing ecosystem serving **15M+ users** in Saudi Arabia.

05 **10+ apps** published on the App Store, used by millions across MENA.

06 **30+ projects** delivered across a 12-year career.

07 Early Swift adopter (WWDC 2014) and early SwiftUI adopter (WWDC 2019) — a habit of betting on the right platform shift early.

FULL-STACK ENGINEER · BACKEND · WEB · MOBILE · 12+ YEARS   SET IN FRAUNCES & JETBRAINS MONO
EDITION 2026 · CAIRO → RIYADH